Normalising Flows

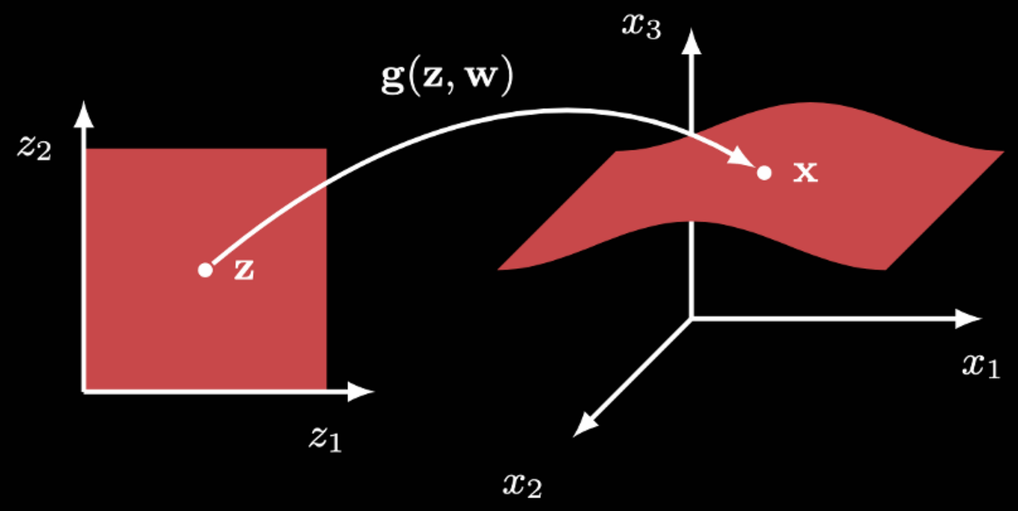
Lecture 8

Ojas Shirekar | 21st May 2026

Recap: VAE

Posterior and Likelihood Approximation

$$p(\mathrm{x} \mid \mathrm{z}, \mathrm{w}) = \mathcal{N}\left(\mathrm{x} \mid \mathrm{g}(\mathrm{z}, \mathrm{w}), \sigma^2 \mathrm{I}\right)$$



Intractable likelihood

$$p(\mathbf{x} \mid \mathbf{w}) = \int p(\mathbf{x} \mid \mathbf{z}, \mathbf{w}) p(\mathbf{z}) d\mathbf{z}$$

And so you decided to use variational approximation and amortisation.

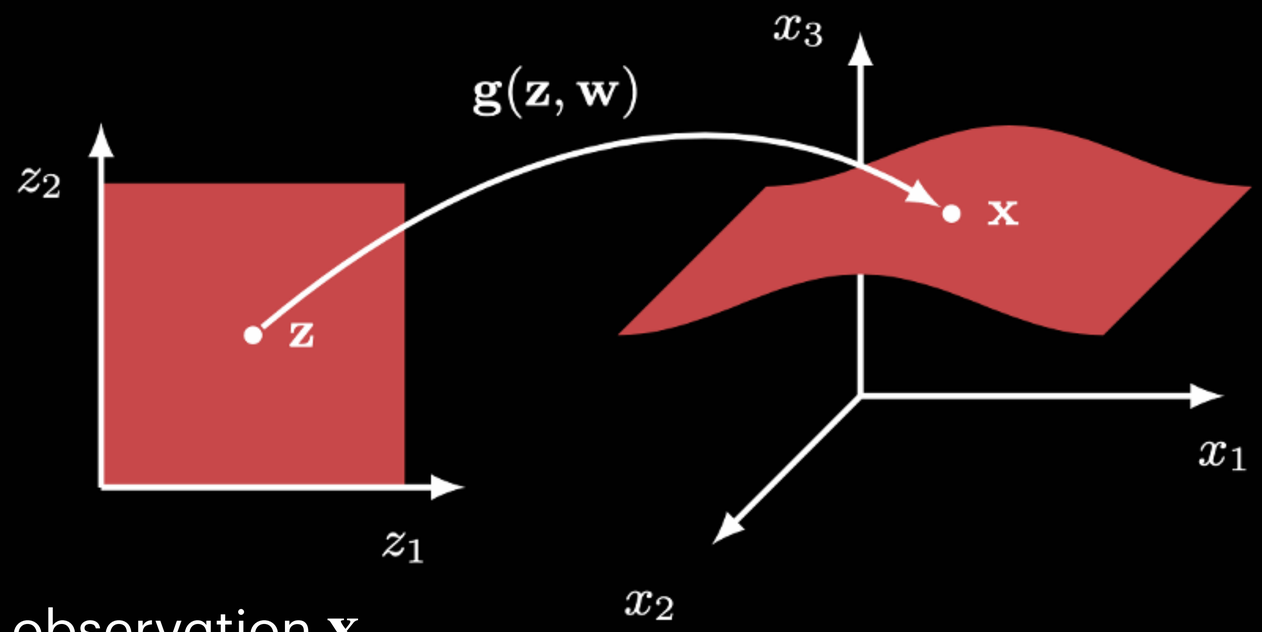
$$\log p(\mathbf{x}|\mathbf{w}) \geq \mathbb{E}_{q(\mathbf{z}|\mathbf{x},\boldsymbol{\phi})}\left[\log p(\mathbf{x}|\mathbf{z}, \mathbf{w})\right] - \mathrm{KL}\left(q(\mathbf{z}|\mathbf{x}, \boldsymbol{\phi}) \,\|\, p(\mathbf{z})\right) = \mathcal{L}(\mathbf{x}; \mathbf{w}, \boldsymbol{\phi})$$

Now, let's rewind a little bit

Recap: VAE

Why was VI used?

$$p(\mathrm{x} \mid \mathrm{z}, \mathrm{w}) = \mathcal{N}\left(\mathrm{x} \mid \mathrm{g}(\mathrm{z}, \mathrm{w}), \sigma^2 \mathrm{I}\right)$$



$g : \mathbf{z}, \mathbf{w} \rightarrow \mathbf{x}$; specifies how to generate the observation $\mathbf{x}$

true posterior $p(\mathbf{z} \mid \mathbf{x})$ - don't know / can't calculate

likelihood – intractable

Because g is a non-linear map

Therefore, you need VI; and cannot do exact MLE

An alternate solution

Normalising Flows

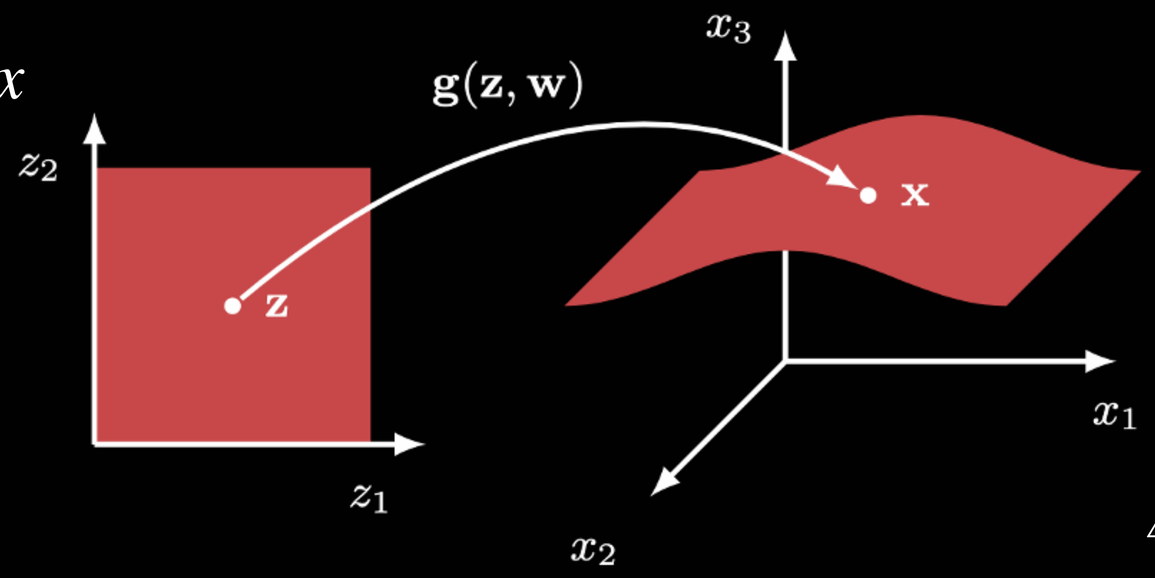
Non-linear g makes posterior intractable; forces us to use VI

What if we enforce g to be an invertible class of functions?

$$x = g(z, w), \qquad z = g^{-1}(x, w)$$

This way, no posterior calculation + bijective $z \mapsto x$

But how?



Normalising Flows

Change of Variables/Measures

$\mathbf{z}$ has a known density $p_{\mathbf{Z}}(\mathbf{z})$

$\mathbf{x}$ is obtained by an *invertible*, *differentiable* map $\mathbf{x} = g(\mathbf{z})$; so $\mathbf{z} = g^{-1}(\mathbf{x})$

Then density of $\mathbf{x}$ is:

$$p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{Z}}\left(g^{-1}(\mathbf{x})\right) \left|\det J_{g^{-1}}(\mathbf{x})\right|$$

Normalising Flows

Change of Variables/Measures - Jacobian

$$p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{Z}}\left(g^{-1}(\mathbf{x})\right) \left| \det J_{g^{-1}}(\mathbf{x}) \right|$$

What's a Jacobian?

g is vector valued map $g : \mathbb{R}^d \to \mathbb{R}^d$

Then the Jacobian matrix at point $\mathbf{z}$ is

$$J_g(z) = \frac{\partial g}{\partial z} \rightarrow J_g(\mathbf{z}) = \begin{bmatrix} \frac{\partial g_1}{\partial z_1} & \cdots & \frac{\partial g_1}{\partial z_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_d}{\partial z_1} & \cdots & \frac{\partial g_d}{\partial z_d} \end{bmatrix}$$

Jacobian matrix at point $\mathbf{z}$

Normalising Flows

Change of Variables/Measures - Jacobian

How to *feel* it?

Take a tiny step $\Delta\mathbf{z}$; Jacobian tells you matching step in $\mathbf{x}$:

$$g(\mathbf{z} + \Delta\mathbf{z}) \approx g(\mathbf{z}) + J_g(\mathbf{z})\Delta\mathbf{z}$$

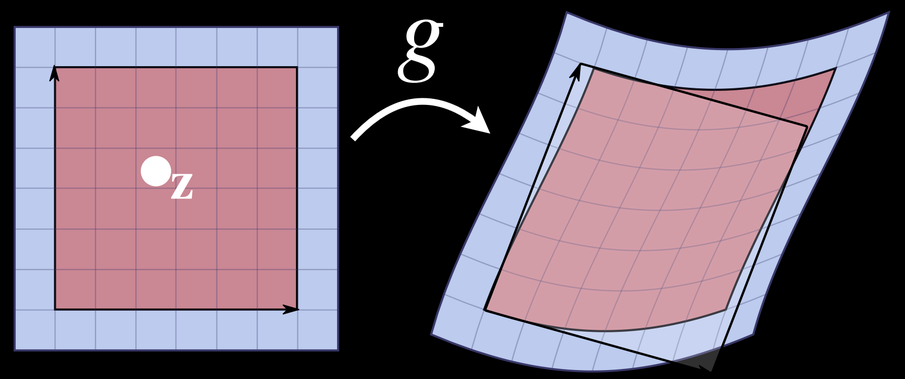
So, J_g is the map's linear approximation!

$\left|\det J_g(\mathbf{z})\right|$ is the volume scaling factor:

- > 1; local expansion
- < 1; local compression
- $= 1$; local volume preservation

$$J_g(\mathbf{z}) = \begin{bmatrix} \frac{\partial g_1}{\partial z_1} & \cdots & \frac{\partial g_1}{\partial z_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_d}{\partial z_1} & \cdots & \frac{\partial g_d}{\partial z_d} \end{bmatrix}$$

Jacobian matrix at point $\mathbf{z}$



First, in 1D

Change of Variables in 1D

How to *feel* it?

$$x = g(z) = 2z$$

A tiny interval around z, dz

Gets stretched to $dx = |g'(z)|\ dz$

$$p_Z(z)dz = p_X(x)dx$$

$$p_X(g(z)) = \frac{p_Z(z)}{\left|g'(z)\right|}$$

$$dx = 2dz$$

$$p_X(x) = \frac{1}{2}p_Z(z)$$

First in 1D

Why determinant?

$$x = g(z)$$

$$g(z + dz) \approx g(z) + g'(z)dz$$

Gets stretched to $dx = |g'(z)|\ dz$

So the image interval as width: $dx \approx |g(z + dz) - g(z)|$

Substitute the approximation: $dx \approx |g'(z)dz|, \qquad dx = |g'(z)|dz$

In 1D, Jacobian is a 1×1 matrix: $J_g(z) = \left[g'(z)\right], \quad \det J_g(z) = g'(z)$

1D: $|g'(z)|$ scales length

2D/3D: $|\det J_g(z)|$ scales area/volume

Normalising Flows

Why do we need a Det of Jacobian?

Mass in a tiny region around $z = p_Z(z)\ dz$

After mapping $x = g(z)$

The same tiny region is stretched/compressed into a region around x

$$dx = \left|\det J_g(z)\right| dz$$

Must obey *mass conservation*

$$p_Z(z)\ dz = p_X(x)\ dx$$

$$p_X(g(z)) = p_Z(z)\left|\det J_g(z)\right|^{-1}$$

$$J_g(\mathbf{z}) = \begin{bmatrix} \frac{\partial g_1}{\partial z_1} & \cdots & \frac{\partial g_1}{\partial z_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_d}{\partial z_1} & \cdots & \frac{\partial g_d}{\partial z_d} \end{bmatrix}$$

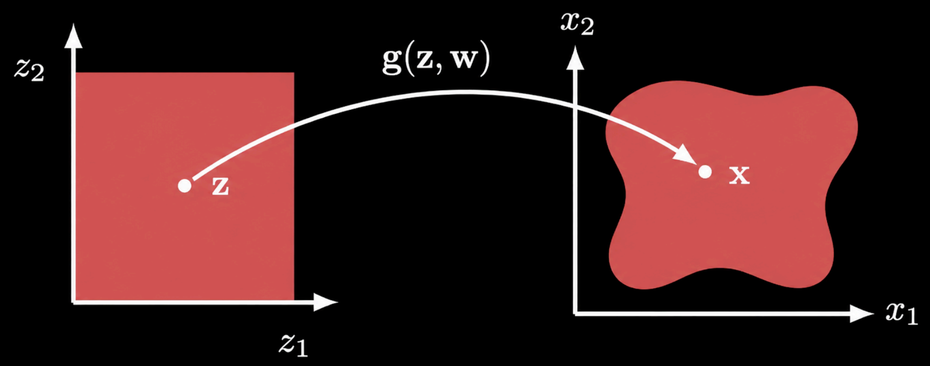
Jacobian matrix at point $\mathbf{z}$

Normalising Flows

Still, why is there a det?

$J_{g^{-1}}$ measures how a tiny volume around $\mathbf{x}$ is stretched

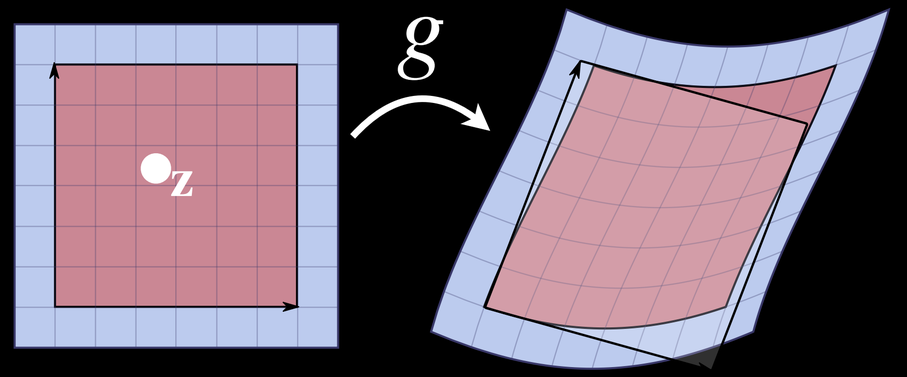
Or squashed when we pull it back to $\mathbf{z}$



In 2D, zoom in on a tiny **square** around $\mathbf{z}$

$\mathbf{e}_1 = \begin{bmatrix}1\\0\end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix}0\\1\end{bmatrix}.$ ← 2D unit basis vectors

Push its corners through g you get a little **parallelogram**



$\mathbf{a} = J_g\mathbf{e}_1 =$ first column , $\mathbf{b} = J_g\mathbf{e}_2 =$ second column

Normalising Flows

Change of Variables/Measures

$$\mathbf{e}_1 = \begin{bmatrix}1\\0\end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix}0\\1\end{bmatrix}. \qquad \leftarrow \text{2D unit basis vectors}$$

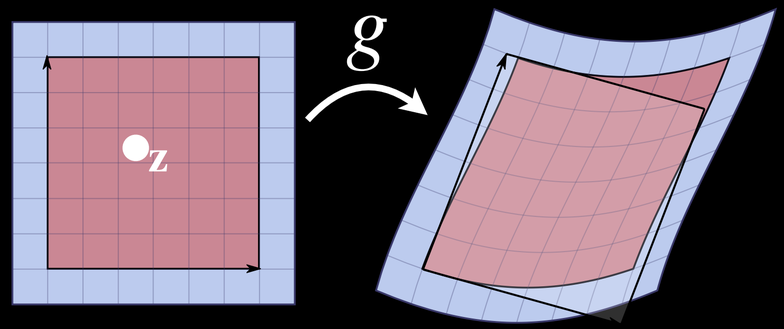
$$\mathbf{a} = J_g \mathbf{e}_1 = \text{ first column }, \quad \mathbf{b} = J_g \mathbf{e}_2 = \text{ second column}$$

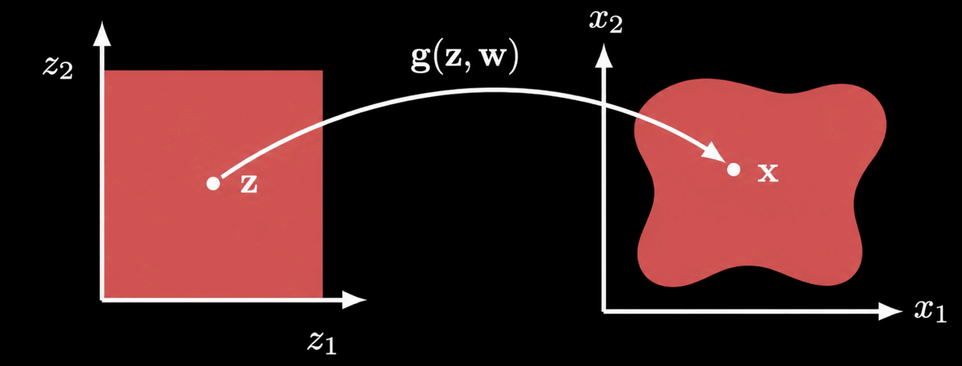
$$J_g(\mathbf{z})\mathbf{e}_1 = \begin{bmatrix}a & b\\c & d\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}a\cdot 1 + b\cdot 0\\c\cdot 1 + d\cdot 0\end{bmatrix} = \begin{bmatrix}a\\c\end{bmatrix}$$

For any 2 vectors, area of ||-gram is: $\left|\det[\mathbf{a} \quad \mathbf{b}]\right|$

Area of new shape, after warp $= \left|\det J_g(\mathbf{z})\right| \times 1 \text{ unit}^2$

$$p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{Z}}\left(g^{-1}(\mathbf{x})\right)\left|\det J_{g^{-1}}(\mathbf{x})\right|$$





Normalising Flows

Change of Variables/Measures

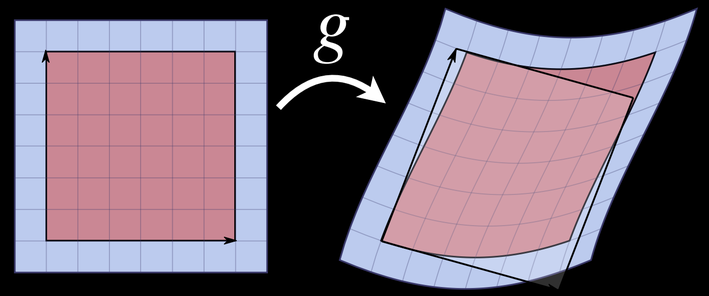
❗ Recall one of the key constraints when moving mass:

$$\int_R p_{\mathbf{Z}}(\mathbf{z})d\mathbf{z} = \int_{g(R)} p_{\mathbf{X}}(\mathbf{x})d\mathbf{x} = 1$$

$$p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{Z}}\left(g^{-1}(\mathbf{x})\right)\left|\det J_{g^{-1}}(\mathbf{x})\right|$$

Before mapping: small volume $d\mathbf{z}$; mass $p_{\mathbf{Z}}(\mathbf{z})d\mathbf{z}$

After: stretched/rotated, whose volume: $d\mathbf{x} = \left|\det J_g(\mathbf{z})\right| d\mathbf{z}$



Mass must be unchanged: $p_{\mathbf{X}}(\mathbf{x})d\mathbf{x} = p_{\mathbf{Z}}(\mathbf{z})d\mathbf{z}$

$$\mathbf{x} = g(\mathbf{z})$$

$$p_{\mathbf{X}}(\mathbf{x})\ d\mathbf{x} = p_{\mathbf{Z}}(\mathbf{z})\ d\mathbf{z}$$

$$p_{\mathbf{X}}(\mathbf{x})\left|\det J_g(\mathbf{z})\right| d\mathbf{z} = p_{\mathbf{Z}}(\mathbf{z})d\mathbf{z}$$

$$p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{Z}}(g^{-1}(\mathbf{x}))\left|\det J_{g^{-1}}(\mathbf{x})\right|$$

Normalising Flows

Training

Choose simple prior distribution as usual, $p(\mathbf{z}) \sim \mathcal{N}(0, I)$

We saw data-density given by: $p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{Z}}(g^{-1}(\mathbf{x})) \left| \det J_{g^{-1}}(\mathbf{x}) \right|$

Take logs: $\log p_X(x) = \underbrace{\log p_Z(z)}_{\text{base density}} + \log \left| \det J_{g^{-1}}(x) \right| .$

Self-study

Training – multiple layers

Compose K simple, bijective maps: $g_\theta = g_K \circ g_{K-1} \circ \cdots \circ g_1, \quad \mathbf{x} = g_\theta(\mathbf{z}), \mathbf{z} = g_\theta^{-1}(\mathbf{x})$

$$p_\theta(x) = p_{\mathbf{Z}}\left(h_K\right) \prod_{k=1}^{K} \left| \det J_{g_k^{-1}}\left(h_{k-1}\right) \right|, \quad h_K = g_\theta^{-1}(x)\,.$$

$$\log p_\theta(x) = \underbrace{\log p_{\mathbf{Z}}\left(g_\theta^{-1}(x)\right)}_{} + \sum_{k=1}^{K} \log \left| \det J_{g_k^{-1}}\left(h_{k-1}\right) \right|$$

Step	**Current variable**	**Density so far**
0	$h_0 = x$	$p_\theta(h_0)$
1	$h_1 = g_K^{-1}(h_0)$	$p_\theta(h_0) = p_{K-1}(h_1)\,\lvert\det J_{g_K^{-1}}(h_0)\rvert$
2	$h_2 = g_{K-1}^{-1}(h_1)$	$p_{K-1}(h_1) = p_{K-2}(h_2)\,\lvert\det J_{g_{K-1}^{-1}}(h_1)\rvert$
$\vdots$	$\vdots$	$\vdots$
K	$h_K = g_\theta^{-1}(x) = z$	$p_1(h_{K-1}) = p_0(h_K)\,\lvert\det J_{g_1^{-1}}(h_{K-1})\rvert$

Recap

- Normalising Flows
  - Jacobian and its interpretation
  - mini-derivation
- Next Lecture: Flow Matching

Advanced Topics

- RealNVP and co.
- Conditional Normalising Flows
- Flow Matching

References

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- Computational Optimal Transport – (https://arxiv.org/abs/1803.00567)
- Optimal Transport for Machine Learners – (https://arxiv.org/abs/2505.06589)
- The Elegance of Optimal Transport – Hugo Cisneros –(https://hugocisneros.com/blog/the-elegance-of-optimal-transport/)
- Flows – (https://jmtomczak.github.io/blog/3/3_flows.html)
- 3.8: Jacobians – Mathematics LibreTexts
- The Jacobian of a Transformation